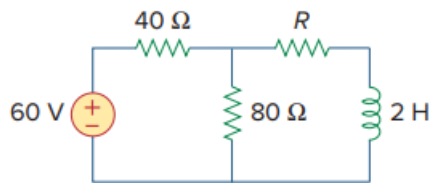


Problem

In the circuit of Fig. 7.101, find the value of R for which the steady-state energy stored in the inductor will be 2 J.

Fig. 7.101:



Step-by-step solution

Step 1 of 3

Assume that the current flowing through the inductor is i .

Consider that the energy stored in inductor 2 J.

$$W = \frac{1}{2} Li^2$$

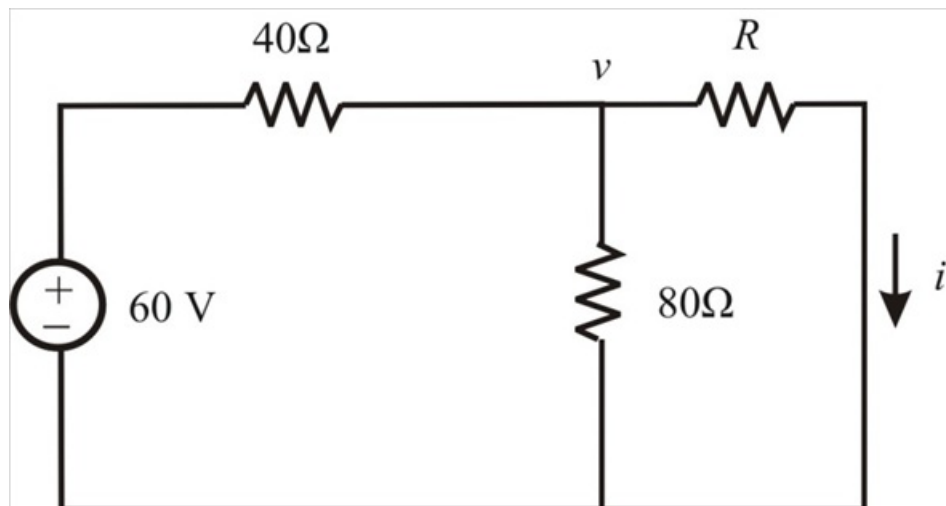
$$2 = \frac{1}{2} \times 2 \times i^2$$

$$i^2 = 2$$

$$i = 1.414 \text{ A}$$

Step 2 of 3

Consider that the inductor acts like a short circuit in the steady state condition. Redraw the circuit as shown in the following figure:



Step 3 of 3

Apply the Kirchhoff's current law at the node of voltage v .

$$\frac{v-60}{40} + \frac{v}{80} + i = 0$$

$$\frac{v-60}{40} + \frac{v}{80} + 1.414 = 0$$

$$v\left(\frac{1}{40} + \frac{1}{80}\right) = \frac{60}{40} - 1.414$$

$$v = 2.293 \text{ V}$$

Apply the ohms law.

$$v = iR$$

$$2.293 = 1.414 \times R$$

$$R = 1.62 \, \Omega$$

Therefore, the value of unknown resistor is $1.62 \, \Omega$.